SHUOFEI ZAN

Xi'an, China | shuofeizan@gmail.com | Research portfolio: github.com/Adele0302/shuofei-zan-design-portfolio

RESEARCH INTERESTS

Creativity-support technologies and human–AI co-creation; evaluating how computational guidance shapes exploration, autonomy, engagement and positive experience in creative work.

EDUCATION

Shaanxi University of Science and Technology Sep 2024 – Jul 2027 (expected)
Master's Degree Candidate in Design | Full-time | GPA 3.65/5.0 | Average 86.53/100

Shaanxi University of Science and Technology Sep 2019 – Jun 2023
BEng in Industrial Design | GPA 3.39/5.0 | Average 84.77/100

Ulster University | Joint programme with SUST Awarded Jun 2023
BSc (Hons) in Design Engineering | First Class Honours

SELECTED PUBLICATIONS

- Sang, Z., **Zan, S.**, Li, L., Fu, X., Liu, Z., Chen, D., Yuan, X., & Shi, P. (2025). Comparative study on the color characteristics of sky-green porcelain in the four kilns of a Ru kiln system from the Northern Song Dynasty. Open Ceramics, 23, 100821. https://doi.org/10.1016/j.oceram.2025.100821
- Sang, Z., Chen, D., Wang, X., **Zan, S.**, Yuan, X., Sang, Y., & Fu, X. (2026). Authenticity identification of the "golden threads and iron wires" patterns of Southern Song Ge ware [English translation; article in Chinese]. China Ceramics, 62(5), 102–112. https://doi.org/10.16521/j.cnki.issn.1001-9642.2026.05.012

RESEARCH EXPERIENCE

Interactive Guidance and Behavioural Evaluation **2025–2026**

- Built a browser-based within-subject study with 18 participants and three guidance conditions: unguided design, rule cues, and refinement cues with target information.
- Logged recommendation acceptance, rejection, undo, revisions and task time to examine both design outcomes and how people respond to computational suggestions.
- Guided conditions took longer on average (75.9–77.6 s vs 53.3 s unguided); rule-distance differences were not statistically significant, and recommendation acceptance was 58.3% in each guided condition.

Interpretable Ceramic Form Modelling and Preference Evaluation **2024–2026**

- Quantified 225 ancient and modern vessel outlines (165 ancient; 60 modern), standardising contour descriptions for statistical comparison.
- PCA explained 85.12% in PC1 and 93.24% cumulatively in PC1–PC2; compared Ridge and RBF-SVR preference models using cross-validated rank correlation.
- Translated observed model relationships into five interpretable design-proportion rules that can be inspected and tested as design guidance.
- Validated generated forms in a separate 30-participant, 48-question 2AFC study: rule-guided forms beat weaker baselines, but not highly rated real examples.

HoberGlow — Parametric Mechanism and Design Prototyping **2025–2026**

- Modelled a transformable scissor-linkage luminaire in Rhino/Grasshopper, exploring kinematic constraints and opening/closing configurations.
- Used a reduced Karamba3D structural proxy and exploratory optical comparisons to compare configurations and document modelling assumptions.
- Documented motion states and design alternatives to communicate mechanism constraints and prototype trade-offs.

Ceramic Materials, Glaze Colour and Heritage Evidence 2024–2026

- Contributed to collaborative studies of historical Chinese ceramics, interpreting XRF, XRD, Raman, SEM-EDS and XPS evidence as relevant to each research question.
- Connected compositional, phase and microstructural evidence with visual comparison across Ru-system, Longquan, Yaozhou and Ge ware; also examined Tang-period yellow glazes and Jin-period moon-white glaze.
- Linked measurement provenance and visual comparison to evidence-based interpretations of glaze colour and authenticity; reconstructed 60 Ru- and Yaozhou-related forms in Rhino.

Urban Second Foyer — User-centred Service Design 2026

- Used stakeholder analysis, user journeys and service blueprinting to translate outdoor workers' rest needs into a public-space support concept.

Rest Light — Affective Interaction Concept 2026

- Developed an early-stage AI-mediated emotional-support concept using product form, ambient lighting and interaction scenarios.

Airport Tower Operator Seat — Ergonomic Design 2022–2023

- Designed an operator seat and workstation concept for airport tower work as the undergraduate thesis, translating a specialised work context into a product-design proposal.

Shared Kitchen — Community Service Design 2025–2026

- Used field research, stakeholder mapping, personas, user journeys and a service blueprint to develop a shared-kitchen service concept.

AWARDS & SCHOLARSHIPS

Scholarship | 2024–2025; Scholarship | 2020–2021

China Good Creativity Competition, Shaanxi Division — First Prize (Urban Second Foyer) and Third Prize (Rest Light), 2026.

Huacan — Regional Third Prize (2025); NCDA, Shaanxi Division — Third Prize (2025).

METHODS AND TECHNICAL COMPETENCIES

Human-centred research: 2AFC preference evaluation, within-subject study design, browser-based interaction logging, acceptance/rejection, undo/revision and task-time measures.

Computational design: Python, PCA, Ridge/RBF-SVR, Rhino/Grasshopper, Karamba3D for exploratory structural analysis, and Unity prototyping.

Materials and heritage: interpretation of XRF/XRD/XPS/Raman/SEM-EDS evidence, quantitative colour and morphology analysis, and scientific visual communication.

Introductory training in eye-tracking, EEG and electrodermal activity (EDA/GSR) methods.

LANGUAGES AND RESEARCH LINKS

Chinese (native) | English (IELTS Academic overall 6.5; all components ≥ 6.0).

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